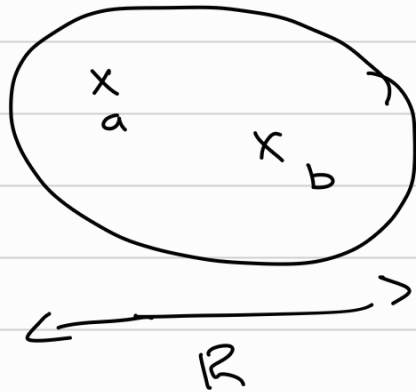


* There are no local observables:

- (i) ∞ - from systematic errors from Quantum corr. from apparatus in the System of interest.
- (ii) ∞ - from having to infinitely do the experiment ∞ -times to narrow the result...
i.e. get a convergent value.

$\leadsto$ For an experiment done in a "Room" of size R



B. H. Entropy



Then, due to gravity, $\exists$ an error $e^{-S_{\text{rm}}}$

$$S_{\text{rm}} = R^2 / G \hbar = M_{\text{Pl}}^2 R^2$$

* The error we find is there for:

$$\sim e^{-M_{\text{Pl}}^2 R^2}$$

• "Retreat" to Boundary to make measurements

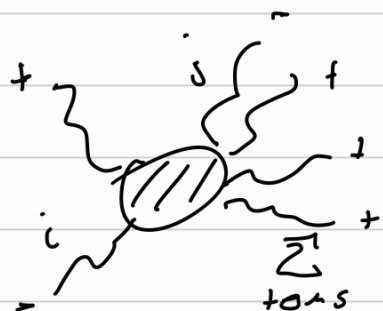
I: There are no local observables in spacetime (i.e. No real measurement can be done w/o the influence of gravity).

B.H. info: Can only info is lost if we have measurements of the order $\sim S_{\text{B.H.}}^{-2}$

* AdS/CFT: doesn't tell us how to ~~understand~~ ^{think abt} emergent space-time.

→ Time is same in AdS as in the Boundary.

* local observables don't exist...



2 gluons $\rightarrow 4$

220 Feynman Diagrams

$$\sum_{\text{loops}} = \langle ij \rangle^4 / \langle 12 \rangle \langle 23 \rangle \dots \langle 61 \rangle$$

→ Newton $\rightarrow$ Prin. of least Action

F.D's $\rightarrow$ This formalism

F.D's: Locality & Unitarity is Manifest

Scattering Amp: $M^{h_1 \dots h_n}(p_1, \dots, p_n)$ h_i : helicity

Lorentz trans. on p_i

$$M^{h_1 \dots h_n}(\Lambda p_1, \dots, \Lambda p_n) = e^{i\Theta(\Lambda, p)} M^{h_1 \dots h_n}(p_1, \dots, p_n)$$

• In QFT: Polarization vectors $\epsilon_{\mu}^{h_i}$

$$\epsilon_{\mu_1}^{h_1} \dots \epsilon_{\mu_n}^{h_n} M^{\mu_1 \dots \mu_n}(p_1, \dots, p_n)$$

$$\epsilon_{\mu}(p) \rightarrow (\Lambda \epsilon)_{\mu}(p) + \alpha p_{\mu}$$

• Lorentz in constraints: $p_{\mu} \epsilon^{\mu} = 0$
leaves 3 D.O.F.

Can't get rid of



ex: Take $\epsilon_{\pm}^{\mu} = (0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0)$, $p^{\mu} = (1, 0, 0, 1)$

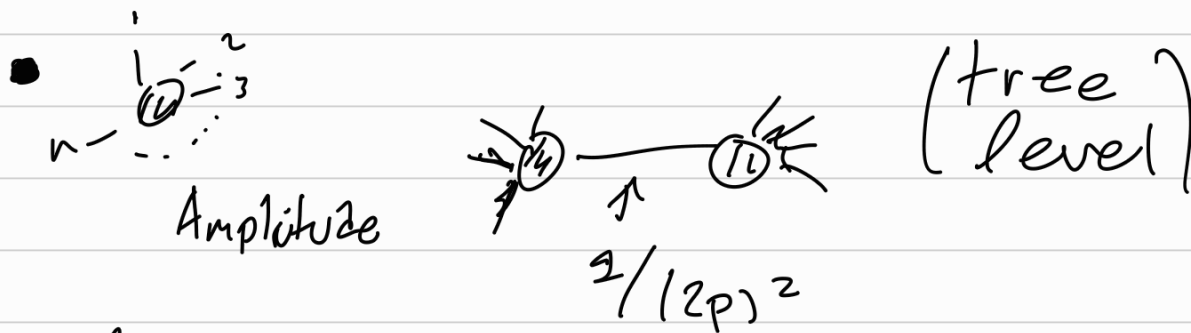
• Identify: $\epsilon^{\mu} \sim \epsilon^{\mu} + \alpha(p) p^{\mu}$

Gauge redundancy

$$P_\mu M^\mu(p_1, \dots, p_n) = 0$$

→ Redundancy manifest due to our redundancy.
Redundant → keep locality

* F.D's carry redundant Gauge information



Locality ~~and~~ singularities of Amplitudes

• Nonlocal Poles: $\sim 1/(p^2 \partial^2 R^2 T)$

• Gaining info by losing locality.

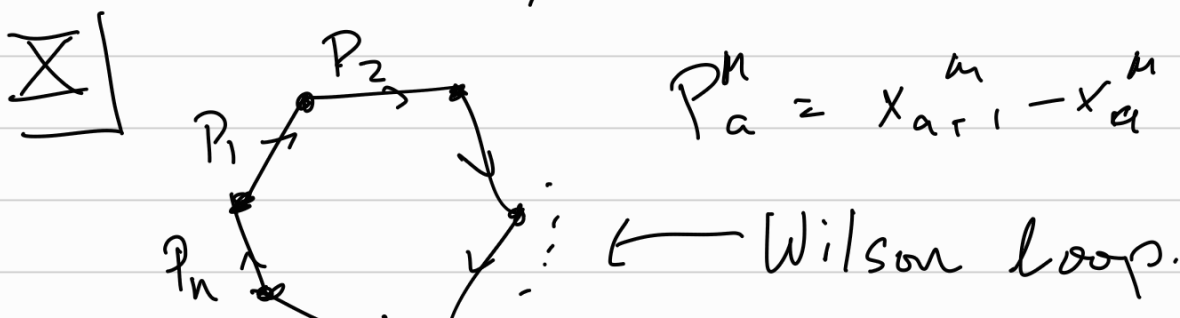
* $N=4$ SYM:

• Conformal symm: "Dual" conf. symm.

AdS/CFT: type of T-duality for strong coupling of $N=4$ SYM.

$$\sum_{A \supset \text{generators}} \text{Tr}(T^{a_1} \dots T^{a_n}) M(a_1, \dots, a_n)$$

$M(1, \dots, n)$: cyclic



- Dual Space $\mathbb{X}$:
Conformal symm on $\mathbb{X}$ space
→ M invariant under $\text{Conf}(\mathbb{X})$

* QCD is Conf. invariant in the Dual space.
• Not seen in the Lagrangian formulation.

- "T-duality" does not commute w/ other symmetries.

→ Symm. Acts on all n -particles and NOT on single states.

* locality does not allow for us to see this...

(Yangian Algebra) i.e. Yangian invariance.